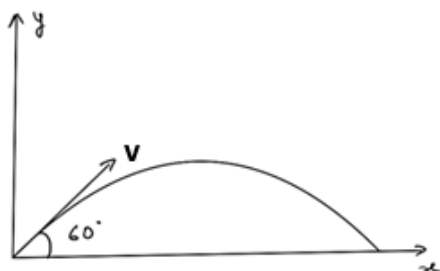


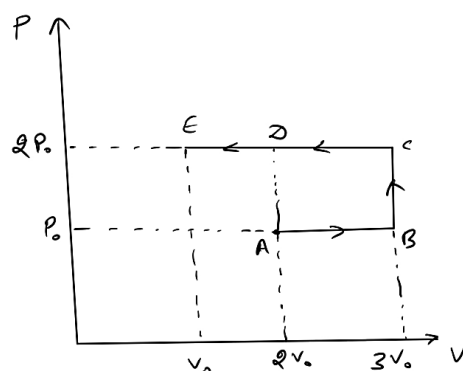
PHYSICS

- Q1** A particle is projected with kinetic energy K . Find the kinetic energy of the particle at highest point of its trajectory.

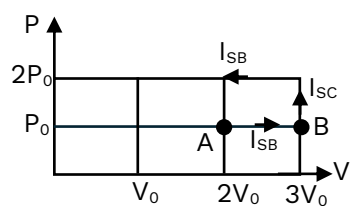


Sol. $k = \frac{1}{2}mv^2$, $\frac{V}{2} = \frac{k}{4}$

- Q2** Find total work done from A to E



- (1) $-3P_0V_0$
 (2) $3P_0V_0$
 (3) $2P_0V_0$
 (4) $5P_0V_0$



Sol.

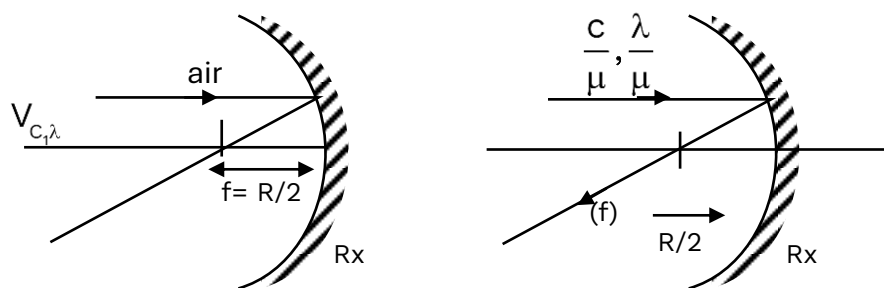
$$W_{AB} = P\Delta V = P_0 (3V_0 - 2V_0) = P_0V_0$$

$$W_{BC} = 0$$

$$W_{CD} = 2P_0 (V_0 - 3V_0) = -4P_0V_0$$

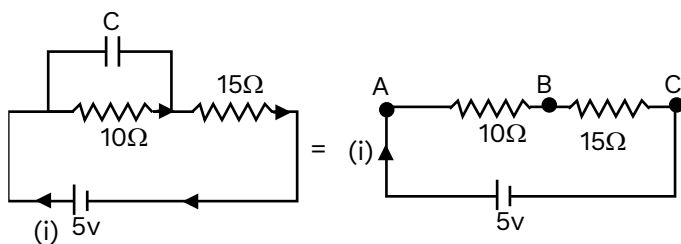
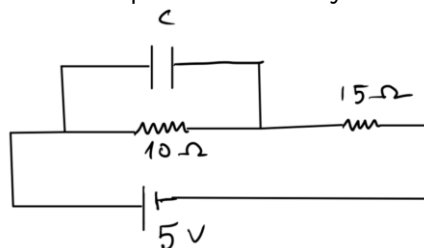
$$W_{net} = P_0V_0 + 0 - 4P_0V_0 = -3P_0V_0$$

- Q3** A concave mirror whose focal length is f which is dipped in a medium whose refractive index is μ then focal length of the mirror in the medium will be :



Sol.

- Q4** In the given RC circuit, Charge on the capacitor at steady state will be:-



Sol.

$$R_{eq} = 25\Omega$$

$$Q = CV_{AB}$$

$$i = \frac{V}{R_{eq}} = \frac{5}{25}$$

$$Q = C \times 2$$

$$Q = 2C$$

$$i = \frac{1}{5} \text{ Amp.}$$

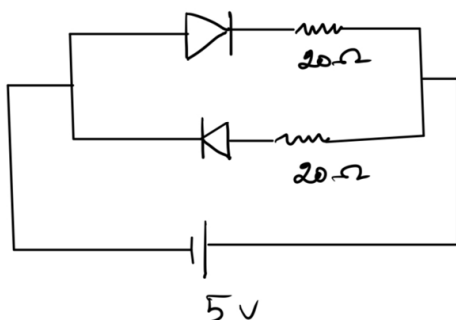
$$V_{AB} = i \times R_1 = \frac{1}{5} \times 10 = 2 \text{ volts}$$

- Q5** A light of wavelength ' λ ' is incident on a metal having work function $\phi = 3.4 \text{ eV}$. The stopping potential measured for the photoelectric current setup is 1.6 eV find the value of λ [$hc = 12400 \text{ eV \AA}$]

Sol. Stopping potential = max KE
max KE = 1.6 eV
E of photons = $3.4 + 1.6 = 5 \text{ eV}$

$$\lambda = \frac{12400}{5} = 2480 \text{ \AA}$$

Q6 Find in the current circuit



Sol. Only one diode is forward biased

$$I = \frac{5}{20} = \frac{1}{4} \text{ A}$$

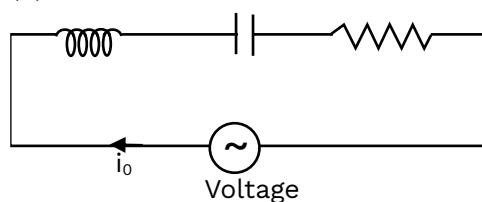
Q7 In a series LCR circuit, inductance $L = 100 \text{ mH}$, and resistance $R = 300 \Omega$ and capacitance $C = 2.5 \mu\text{F}$. The angular frequency of the source when current has maximum amplitude in the circuit is

(1) $\frac{10^4}{2\pi} \text{ rad/s}$

(2) $\frac{10^5}{2\pi} \text{ rad/s}$

(3) 10^5 rad/s

(4) 10^6 rad/s



Sol.

$$V_0 = \text{fix}$$

$$\text{Max } x_0 = \frac{V_0}{Z_{\min}}$$

$$Z_{\min}$$

$$X_L = X_C$$

$$L_{\omega} = \frac{1}{L_{\omega}}$$

$$\omega^2 = \frac{1}{L_C} = \frac{1}{100 \times 10^{-3} \times 2.5 \times 10^{-6}}$$

$$\omega^2 = \frac{1}{25 \times 10^{-8}}$$

$$\omega = \frac{1}{5 \times 10^{-4}} = 2000$$

$$\omega^2 = \frac{1}{L_C}$$

$$= \frac{1}{100 \times 10^{-3} \times 2.5 \times 10^{-6}}$$

$$= \frac{1}{250 \times 10^{-9}}$$

$$\omega^2 = \frac{1}{25 \times 10^{-8}}$$

$$\Rightarrow \omega = 2000$$

- Q8** For the prism with refracting angle A and refractive index $\mu = \sqrt{3}$. The minimum deviation is A . Find the value of A .

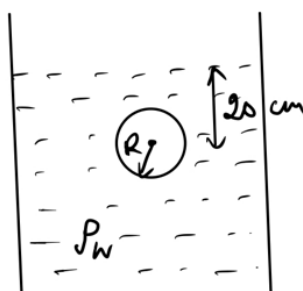
Sol.
$$\mu = \frac{\sin\left(\frac{A + \delta_{\min}}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$= \frac{\sin(A)}{\sin(A/2)}$$

$$\frac{\sqrt{3}}{2} = \cos\left(\frac{A}{2}\right)$$

$$A = 60^\circ$$

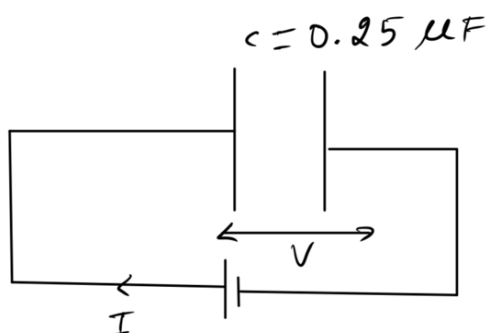
- Q9** Find pressure inside the bubble of radius $R = 1$ mm. If atmospheric pressure is 10^5 N/m² and density of water $\rho_w = 10^3$ kg/m³.

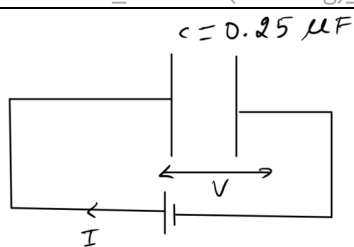


Sol.
$$10^5 + \rho_w gh + \frac{2T}{R} = P_{\text{inside}}$$

$$P_{\text{inside}} = 10^5 + 2 \times 10^3 + 180$$

- Q10** Find rate of change of voltage $\frac{dv}{dt}$, given $I = 0.25$ mA.



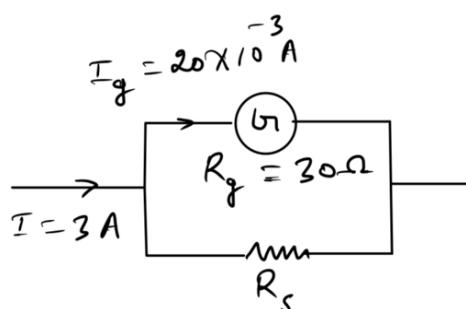
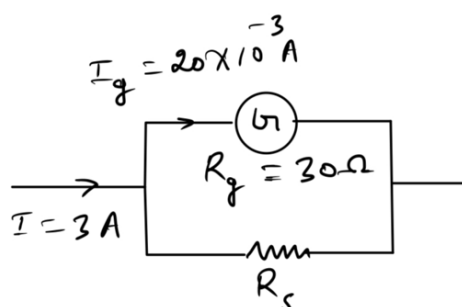
**Sol.**

$$Q = CV$$

$$\frac{dQ}{dt} = C \frac{dv}{dt}$$

$$\frac{dv}{dt} = 1000$$

Q11 The value of shunt resistance (R_s) = $\frac{30}{x}$. Find the value of x.

**Sol.**

These 2 resistance will be in parallel

$$I_g R_g = (I - I_g) R_s$$

$$20 \times 10^{-3} \times 30 = (3 - 20 \times 10^{-3}) R_s \quad \frac{6}{30} = \frac{30}{x}$$

$$X = 150$$

Q12 In a relation $E = \alpha^3 e^{-\beta t}$, percentage error of α is 1.2% and time is 1.6 %. Find % error in measurement of E. at $t = 5$ sec ($\beta = 0.3$)

Sol. $E = \alpha^3 e^{-\beta t}$

$$\frac{\Delta E}{E} \times 100 = ?$$

$$\log E = 3 \log \alpha - \beta t$$

$$\left(\frac{\Delta E}{E} = +3 \frac{\Delta \alpha}{\alpha} - \beta \Delta t \right) 100$$

$$\frac{\Delta E}{E} \times 100 = \frac{3 \Delta \alpha}{\alpha} \times 100 - \beta \times \Delta t \times 100$$

$$\frac{\Delta x}{x} \times 100 = 1.2$$

$$\beta = 0.3$$

$$\frac{\Delta t}{t} \times 100 = 1.6$$

$$\Delta t \times 100 = 1.6 \times 5$$

$$= 8.0$$

$$= 8$$

$$\frac{\Delta E}{E} \times 100 = 3 \times 1.2 - (0.3) \times (-8)$$

$$\frac{\Delta E}{E} \times 100 = 3.6 + 2.4 = 6.0$$

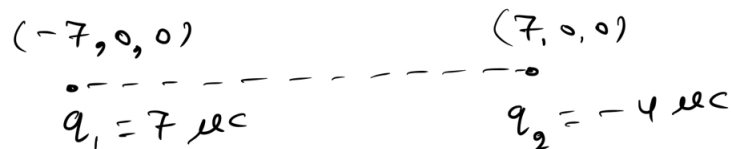
Q13 A travelling wave given by,

$y = A_0 \sin(20 \times 10^{-3}x + 600t)$ where x is in mm. Find the velocity of wave.

Sol. $v = \omega/k$

$$= \frac{600}{20} = 30 \text{ m/s}$$

Q14 In the given two charge particle system, find the potential energy of the system.



Sol. $(-7, 0, 0)$ $(7, 0, 0)$
 $q_1 = 7\mu$ $Q_2 = -4\mu C$

$$E = \frac{-(7) \times 4 \times 9 \times 10^9}{14}$$

$$E = -8 \times 10^9$$

Q15 A disc of radius $R = 1$ m and mass $m = 1$ kg is rotating about fixed axis passing through centre and normal to plane. Find the power delivered by external force at $t = 2$ sec if θ varies as $\theta = 5t^2 - 8t$.

Sol. $\theta = 5t^2 - 8t$

$$\omega = 10t - 8$$

$$\alpha = 10$$

$$\tau = I\alpha$$

$$\tau = 10 \times \frac{1}{2} = 5$$

$$\text{Power} = \tau\omega = 5 \times 12 = 60 \text{ W}$$

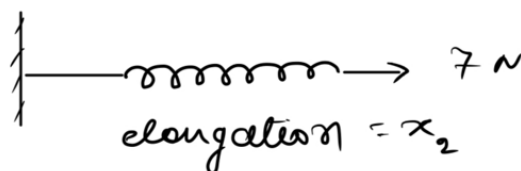
Q16 Statement 1: Graph of frequency f of X ray and atomic number Z of heavy nucleus is straight line, in X ray emission.

Statement 2: Graph of square root of frequency $\sqrt{f}$ of X ray and atomic number Z of heavy nucleus is straight line, in X ray emission.

- (1) Statement 1 is correct and statement 2 is correct
 (2) Statement 1 is incorrect and statement 2 is correct
 (3) Statement 1 is correct and statement 2 is incorrect
 (4) Statement 1 is incorrect and statement 2 is incorrect

Sol. $\sqrt{f} \propto Z$ (Theory based)

Q17 When elongation in the spring is $5x_1 - 2x_2$ then find tension in the spring.



Sol. $x_1 = \frac{5}{K}$

$$x_2 = \frac{7}{K}$$

$$F_s = Kx$$

$$= K (5x_1 - 2x_2) = K \left(5 \times \frac{5}{K} - 2 \times \frac{7}{K} \right)$$

$$= 25 - 14 = 11 \text{ N}$$

Q18 In an electromagnetic wave of frequency $f = 20 \text{ MHz}$, amplitude of electric field $E = 9.3 \times 10^{-4} \text{ N/C}$. Find amplitude of magnetic field. Use $C = 3 \times 10^8 \text{ m/sec}$.

Sol. $B = \frac{E}{C} = \frac{9.3 \times 10^{-4}}{3 \times 10^8}$
 $= 3.1 \times 10^{-4} \text{ T}$.

Q19 In YDSE experiment, the intensity ratio $\frac{I_{\max}}{I_{\min}} = \frac{9}{4}$ and width of one slit is d and other slit is xd . Find the value of x

Sol. intensity $\propto (\text{width})^2$

$$I_2 = kd^2$$

$$I_1 = kx^2d^2$$

$$\frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \frac{9}{4}$$

$$= \frac{\sqrt{l_1} + \sqrt{l_2}}{\sqrt{l_1} - \sqrt{l_2}} = \frac{3}{2}$$

$$= \frac{cd + d}{xd - d} = \frac{3}{2}$$

$$= 2xd + 2d = 3xd - 2d$$

$$x = 5$$

Q20 A satellite of mass $\frac{M}{2}$ is orbiting around earth of mass M at height $h = \frac{R}{3}$ from the surface of earth,

where R is radius of earth. Angular momentum of satellite is $M\sqrt{\frac{GMR}{x}}$. Find the value of x .

Sol. $L = mvr$

$$= \frac{m}{2} \sqrt{\frac{GM}{4 \frac{R}{3}}}$$

$$= \frac{m}{2} \sqrt{Gm \times \frac{4R}{3}}$$

$$= M\sqrt{\frac{GMR}{3}}$$

$$x = 3$$

Q21 Match the column I with Column-II

Column-I

- (A) Magnetic permeability
- (B) magnetic field
- (C) magnetic moment
- (D) Torsional constant

Column-II

- (P) $ML^0T^{-2}A^{-1}$
- (Q) $M^0L^2T^0A$
- (R) $MLT^{-2}A^{-2}$
- (S) ML^2T^{-2}

Sol. $[\mu] = MLT^{-2}A^{-2}$
 $[B] = ML^0T^{-2}A^{-1}$
 $[M] = M^0L^2T^0A$
 $[c] = ML^2T^{-2}$